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[Intervention Review]

Total and partial knee arthroplasty versus non-surgical interventions of the knee for moderate to severe osteoarthritis

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ABSTRACT

Rationale

Total and partial knee arthroplasty (TKA, PKA) are common treatments for moderate to severe knee osteoarthritis. While effective in symptom management, these surgeries carry potential risks, necessitating careful evaluation of benefits and harms. High-quality evidence comparing TKA/PKA to placebo, sham, or non-surgical interventions is essential to support informed decision-making.

Objectives

To assess the benefits and harms of TKA and PKA for people with moderate to severe knee osteoarthritis compared to placebo, sham (efficacy), or non-surgical interventions for the knee (effectiveness).

Search methods

We searched CENTRAL, MEDLINE, Embase, and two trial registers from 2010 to January 2025. We also performed reference and citation checks. Two reviewers independently screened studies, extracted data, and assessed risk of bias and the certainty of evidence.

Eligibility criteria

We included randomised controlled trials of adults with moderate to severe osteoarthritis receiving TKA/PKA compared to placebo, sham, or non-surgical treatments including physiotherapy, weight loss, pharmacotherapy, braces, insoles, and any intraarticular injection. We excluded studies including participants with other joint conditions or having TKA/PKA because of trauma (e.g. fracture), infection, or revision.

Outcomes

We reported on pain, physical function, knee surgery, patient satisfaction with treatment outcomes, health-related quality of life, serious adverse events, and withdrawals due to adverse events. We measured outcomes at short-term (≤ 6 months), intermediate-term (6 to 12 months), and long-term (≥ 1 year) follow-up.

Risk of bias

We used the Cochrane RoB 1 tool to assess risk of bias.

Synthesis methods

We conducted a narrative synthesis, reported on risk ratios (RRs) for dichotomous data and mean differences (MDs) for continuous data with a 95% confidence interval (CI), and used GRADE to assess the certainty of evidence.

Included studies

One randomised controlled trial conducted in Denmark met the inclusion criteria. The study enrolled 100 adults with primary mild to severe knee osteoarthritis (Kellgren-Lawrence score ≥ 2 to 4) and compared TKA followed by a non-surgical programme (exercise, osteoarthritis education, dietary advice, insoles, pain medication) to the same non-surgical programme alone.

Synthesis of results

Pain

Compared to a non-surgical programme, unilateral TKA may reduce pain at a level that is clinically important at one year (MD 17.60, 95% CI 8.25 to 26.95; 1 study, 100 participants; low-certainty evidence). The certainty of evidence was low given the imprecision of results and risk of bias.

Physical function

Compared to a non-surgical programme, unilateral TKA may improve physical function at one year, but the improvement might not be clinically important (MD 12.40, 95% CI 3.06 to 21.74; 1 study, 100 participants; low-certainty evidence). The certainty of evidence was low given the imprecision of results and risk of bias.

Knee surgery

Compared to a non-surgical programme, unilateral TKA may reduce the need for a follow-up knee surgery (revision or subsequent knee surgery and initial knee surgery) at one year (RR 0.04, 95% CI 0.00 to 0.62; 1 study, 99 participants; low-certainty evidence). The certainty of evidence was low given the imprecision of results and risk of bias.

Patient satisfaction with treatment outcomes

The included trial did not assess patient satisfaction with treatment outcomes.

Health-related quality of life

There is probably no clinically important difference in health-related quality of life between unilateral TKA and non-surgical treatment at one year (MD 0.09, 95% CI 0.01 to 0.18; 1 study, 100 participants; moderate-certainty evidence). The certainty of evidence was moderate due to risk of bias and was not downgraded for imprecision because the effect estimates and the CI did not cross the minimal clinically important difference line.

Serious adverse events

Compared to a non-surgical programme, unilateral TKA may increase serious adverse events, but the evidence is very uncertain (RR 4.00, 95% CI 1.79 to 8.94; 1 study, 100 participants; very low-certainty evidence). The certainty of evidence was very low given the imprecision of results, indirectness, and risk of bias.

Withdrawals due to adverse events

There may be no difference in withdrawals due to adverse events between unilateral TKA and non-surgical treatment, but the evidence is very uncertain ($n = 0$ in both groups, Not estimable; 1 study, 100 participants; very low-certainty evidence). The certainty of evidence was very low given the imprecision of results.

Authors' conclusions

Compared to a non-surgical programme alone, TKA followed by a non-surgical programme may reduce pain at a level that is clinically important; may improve physical function at a level that is not clinically important; and may reduce the need for follow-up knee surgery.

There is probably no clinically important difference in health-related quality of life between TKA followed by a non-surgical programme and a non-surgical programme alone.

TKA followed by a non-surgical programme may increase the risk of serious adverse events, but the evidence is very uncertain.

There may be no difference between groups in withdrawals due to adverse events, but the evidence is very uncertain.

The conclusions of this study should be interpreted with caution due to several limitations: the evidence is based on a single study conducted in Denmark; TKA eligibility criteria used by surgeons were not clearly reported; 12% of adults had mild osteoarthritis; adults who reported severe pain in the previous week were excluded; and the included study employed a broad definition of serious adverse events. These factors may affect the reliability and generalisability of the findings.

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Registration

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PLAIN LANGUAGE SUMMARY

What are the benefits and harms of knee replacement surgery for treating knee osteoarthritis compared to non-surgical treatments?

Key messages

- People with mild to severe knee osteoarthritis who have total knee replacement surgery followed by a non-surgical treatment programme may feel a small reduction in pain at a level that offers a real benefit; may have a small improvement in physical function that may not offer a real benefit; and may have a small reduction in the need for a follow-up knee surgery compared to a non-surgical treatment programme alone. There is likely no difference in terms of real benefits between treatment groups in health-related quality of life.
- The effect of total knee replacement surgery followed by a non-surgical treatment programme on the risk of serious unwanted effects and withdrawals due to unwanted effects compared to a non-surgical treatment programme alone is very uncertain.
- Future studies are needed to compare total knee replacement surgery with non-surgical treatments for osteoarthritis. These studies should look at the benefits and risks of treatment and patient satisfaction.

What is knee osteoarthritis?

Knee osteoarthritis is a common and progressing condition that affects the knee joint and area around it. Osteoarthritis causes pain, stiffness, and reduced mobility, which can be mild, moderate, or severe. These symptoms affect the quality of life of millions of people worldwide.

How is knee osteoarthritis treated?

Treatment for moderate to severe knee osteoarthritis includes non-surgical treatments and knee replacement surgery. The usual non-surgical treatment options are education, exercises, physiotherapy, weight loss, medications, and joint injections.

As osteoarthritis progresses, some people may need knee replacement surgery. During knee replacement surgery, the surgeon replaces the surfaces of the knee joint (shinbone, thighbone, and sometimes the kneecap) with artificial parts made of metal, plastic, or ceramic. A plastic spacer is also placed between the shinbone and thighbone to help the joint move smoothly.

There are many treatment options for knee osteoarthritis, and there is no one best option for everyone. To understand the treatment options, people with knee osteoarthritis need clear, trustworthy information about the possible benefits and risks of non-surgical treatment options and knee replacement surgery to be able to choose the treatment that best fits their personal values, goals, and lifestyle.

What did we want to find out?

We wanted to find out which treatment (knee replacement surgery, non-surgical treatments, dummy treatment, or fake treatment) works best for adults with moderate to severe knee osteoarthritis.

We also wanted to find out if adults who had knee replacement surgery, non-surgical treatments, dummy treatment, or fake treatment had any unwanted effects.

What did we do?

We searched for published studies in the last 15 years that looked at knee replacement surgery compared to non-surgical, dummy, or fake treatments in people with moderate to severe knee osteoarthritis. We summarised and compared the results of the studies and rated our confidence in the evidence based on factors such as study methods and sizes.

What did we find?

We found only one study that involved 100 adults with mild to severe osteoarthritis. The study tested total knee replacement surgery followed by a 12-week non-surgical programme compared to the same 12-week non-surgical programme alone. In the group that had total knee replacement surgery followed by a non-surgical programme, 32 out of 50 adults were women (64%), and people were on average 66 years old. In the group that received a non-surgical programme alone, 30 out of 50 adults were women (60%), and people were on average 66 years old.

Main results

Total knee replacement followed by a 12-week non-surgical programme compared to the same 12-week non-surgical programme alone:

- may reduce pain at one year at a level that offers a real benefit;
- may improve physical function at one year, but the improvement might not amount to a real benefit;
- may reduce the need for a follow-up knee surgery (revision or subsequent knee surgery and initial knee surgery);
- probably results in no difference in terms of real benefits between treatment groups in health-related quality of life at one year.

There was no evidence on patient satisfaction with treatment outcomes.

We are very uncertain about the effect of total knee replacement followed by a 12-week non-surgical programme compared to the same 12-week non-surgical programme alone on the risk of serious unwanted effects at one year and withdrawals due to unwanted effects.

What are the limitations of the evidence?

Our confidence in the evidence is only moderate to very low because we found only one small study conducted between 2011 and 2013; it was unclear how doctors decided who could have knee replacement surgery; a small number of people in the study had mild osteoarthritis instead of moderate to severe; and people who had severe pain the week before were not included in the study, which could have influenced the results of the review. Additionally, people in the study were aware of the treatment received, which could have affected the results.

How up-to-date is the evidence?

The evidence is current to January 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Unilateral TKA compared to non-surgical treatment of the knee for mild to severe osteoarthritis

Unilateral TKA compared to non-surgical treatment of the knee for mild to severe osteoarthritis

Patient or population: mild to severe osteoarthritis

Setting: public outpatient clinics in Denmark

Intervention: unilateral TKA

Comparison: non-surgical treatment of the knee

Outcome	Number of participants (studies)	Relative effect (95% CI)	Anticipated absolute effects (95% CI)			Certainty of the evidence	What happens
			Non-surgical treatment	TKA	Difference		
Pain (improvement from baseline) assessed with: KOOS subscale (0 to 100, higher is better) Follow-up: 12 months	100 participants (1 RCT)	-	The mean improvement in the non-surgical treatment group was 17.2 (95% CI 10.4 to 24.1).	The mean improvement in the TKA group was 34.8 (95% CI 28.1 to 41.5).	MD 17.6 higher (95% CI 8.25 to 26.95 higher)	⊕⊕○○ Low ^{a,b}	Unilateral TKA may reduce pain at a level that is clinically important.
Physical function (improvement from baseline) assessed with KOOS activities of daily living scale (0 to 100, higher is better) Follow-up: 12 months	100 participants (1 RCT)	-	The mean improvement in the non-surgical treatment group was 17.6 (95% CI 11.4 to 23.9).	The mean improvement in the TKA group was 30.0 (95% CI 22.7 to 37.2).	MD 12.4 higher (95% CI 3.06 to 21.74 higher)	⊕⊕○○ Low ^{a,b}	Unilateral TKA may improve physical function, but the improvement might not be clinically important.
Need for a follow-up knee surgery (including revision or subsequent operation for the TKA group or initial TKA for the non-surgical group) Follow-up: 12 months	99 participants (1 RCT)	RR 0.04 (0.00 to 0.62)	26 out of 100 participants needed a subsequent surgery in the non-surgical treatment group.	1 out of 100 participants needed a follow-up surgery in the TKA group.	25 fewer per 100 (95% CI 26 fewer to 10 fewer)	⊕⊕○○ Low ^{a,b}	Unilateral TKA may reduce the need for a follow-up knee surgery (revision or subsequent knee surgery and initial knee surgery).
Patient satisfaction with treatment outcomes	Not reported						
Health-related quality of life (mean improvement) assessed with EQ-5D (0 to 1, higher is better) Follow-up: 12 months	100 participants (1 RCT)	-	The mean improvement in the non-surgical treatment group was 0.115	The mean improvement in the TKA group was 0.206 (95% CI 0.141 to 0.270).	MD 0.09 higher (95% CI 0.01 to 0.18 higher)	⊕⊕⊕○ Moderate ^a	There is probably no clinically important difference in health-related quality of life between

			(95% CI 0.063 to 0.166).				unilateral TKA and non-surgical treatment.
SAEs ^c per 100 participants/12 months Follow-up: 12 months	100 participants (1 RCT)	RR 4.00 (1.79 to 8.94)	The number of SAEs in the non-surgical treatment group was 12 .	The number of SAEs in the TKA group was 48 .	36 more (95% CI 21 to 107 more)	⊕○○○ Very low ^{a,b,c}	Unilateral TKA may increase SAEs, but the evidence is very uncertain. ^c
Withdrawals due to adverse events Follow-up: 12 months	100 participants (1 RCT)	-	0 out of 50 participants withdrew due to adverse events in the non-surgical treatment group.	0 out of 50 participants withdrew due to adverse events in the TKA group.	-	⊕○○○ Very low ^{a,d}	There may be no difference between unilateral TKA and non-surgical treatment, but the evidence is very uncertain.

Abbreviations: CI: confidence interval, KOOS: Knee Injury and Osteoarthritis Outcome Score, MD: mean difference, RCT: randomised controlled trial, RR: risk ratio, SAE: serious adverse event, TKA: total knee arthroplasty

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate; the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited; the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate; the true effect is likely to be substantially different from the estimate of effect.

^aDowngraded one level for risk of bias due to lack of blinding of participants.

^bDowngraded one level for imprecision due to not achieving the optimal information size.

^cDowngraded one level for indirectness, as the definition of SAEs in the study was too inclusive. The definition of SAEs included those having the potential to significantly compromise the clinical outcome, result in significant disability or incapacity, requiring inpatient or outpatient hospital care, and those considered to prolong hospital care, to be life-threatening, or to result in death. All events are included, irrespective of attribution.

^dDowngraded two levels for imprecision due to low number of events.

BACKGROUND

Description of the condition

Knee osteoarthritis is a leading cause of disability worldwide, impacting the health of individuals and populations [1, 2]. Internationally, over 263 million adults are affected by knee osteoarthritis as of 2017 [3]. Osteoarthritis is a progressive condition characterised by varying degrees of change in the cartilage, bones, and surrounding joint structures [4]. Distinctive symptoms accompanying osteoarthritis are joint pain and stiffness and a decrease in joint mobility and function. These symptoms result in poor health-related quality of life in adults [5]. Symptoms in osteoarthritis are typically fluctuating but manageable; however, a minority of adults with osteoarthritis can experience worsening symptoms (e.g. persistent knee pain, pain at night), indicating the need for knee arthroplasty [6, 7].

Knee osteoarthritis is classified based on its severity: mild, moderate, and severe. Severity is determined using many methods, including imaging, clinical symptoms, and physical assessment [4, 8, 9, 10]. Radiographically, knee osteoarthritis shows slow evolution of disease over time (3% to 4% per year), with most adults remaining stable [11]. Yet, it is estimated that an average of 50% of adults diagnosed with knee osteoarthritis will undergo total/partial knee arthroplasty (TKA/PKA) during their lifetime [12].

Description of the intervention and how it might work

A broad range of conservative (pharmacological and non-pharmacological) treatments are available to manage milder osteoarthritis symptoms, and a combination of pharmacological and non-pharmacological therapies are recommended as first-line treatment [13]. Typically, principal core treatments include education (including self-efficacy and management skills), physiotherapy (e.g. exercises, stretches, walking aids, thermotherapy, cryotherapy, insoles, braces), weight loss, and pharmacological therapies (topical and oral non-steroidal anti-inflammatory drugs) [5, 14, 15]. Additional treatment options include intraarticular injections (e.g. corticosteroids), and opioids [14, 15].

For moderate to severe osteoarthritis, when conservative treatment fails to alleviate symptoms, evidence-based guidelines recommend total knee arthroplasty (TKA) and partial knee arthroplasty (PKA) as surgical treatment options [16, 17, 18, 19, 20]. TKA and PKA are not advised for milder osteoarthritis, as they result in poor outcomes [21], involve risks [22], and patients with mild symptoms can meaningfully manage their symptoms through conservative treatments [15]. Some international guidelines recommend TKA for adults who have osteoarthritis symptoms that negatively impact their quality of life (e.g. pain, functional limitations), radiographic evidence of osteoarthritis, or fail to have symptom control with conservative treatments [23, 24]. Recent evidence-based clinical practice guidelines recommend considering the perspectives of adults with osteoarthritis and engaging them in the TKA decision-making process by using shared decision-making approaches [15, 24]. However, there is no agreement on the key clinical indications for TKA in osteoarthritis, which contributes to variations in global clinical practice [25].

TKA involves the surgical replacement of the surface of the bones (tibia, femur, and occasionally patella) with synthetic

prostheses (metal, plastic, ceramic) and the insertion of a spacer (plastic) between the tibia and femur [26]. PKA involves the same procedure, except that only one area of the knee (medial, lateral, or patellofemoral) is replaced [27].

By replacing the structure of the knee joint, TKA and PKA result in good pain management and increased physical function following these surgeries, leading to better health-related quality of life [28, 29, 30, 31]. Both surgeries are associated with rare adverse events (e.g. mortality, cardiac events) [32, 33]. However, surgical revision of the prosthesis may be required in the short term (e.g. infection) and long term (e.g. prosthesis survival) [34].

Conservative treatments for osteoarthritis help to maintain overall joint and physical function, support joint unloading, and lessen joint pain [13, 35].

Why it is important to do this review

Although TKA and PKA are widely recommended surgical treatments to manage moderate to severe osteoarthritis symptoms, patient satisfaction rate varies across studies. A 2012 systematic review reported dissatisfaction rates ranging from 10% to 34% [36]; a 2018 review found a broader range of 0% to 35% [37]; while more recent reviews reported narrower estimates of 5% to 16% in 2023 [38] and 8% to 14% in 2025 [39]. Factors contributing to patient dissatisfaction with TKA and PKA vary.

There is a poor association between surgeons' judgement for TKA/PKA and patients' informed preferences for these surgeries [40, 41, 42]. However, most decisions to undergo TKA/PKA are based on surgeons' judgement without considering patients' perspectives [41, 43, 44, 45]. The criteria deemed most important by orthopaedic surgeons when recommending TKA/PKA are radiographic severity and patient-reported pain [46, 47, 48, 49, 50]. However, the criteria influencing patient preference for surgery are the impact of osteoarthritis on quality of life and pain [41, 48, 51, 52, 53].

Patients' preoperative expectations influence the decision to undergo these surgeries [54]. However, patients may have unrealistic expectations of TKA/PKA outcomes, leading to dissatisfaction with the results of surgery [41, 55, 56, 57, 58]. Not all patients will benefit from surgery, and approximately 20% report persistent postoperative knee pain following TKA [36, 59]. Many patients report receiving little information on the possible surgical outcomes and alternative treatment options, which can be considered as an uninformed decision to have TKA/PKA [51, 53, 60].

To be consistent with patient-centred care [61], it is essential that patients are given evidence-based information about the possible benefits and harms of TKA/PKA and alternative treatment options. PKA and TKA are elective procedures in a pool of many alternative treatment options for knee osteoarthritis, with no best choice [35]. As a result, patients need to be aware of all treatment options, and be given the opportunity to discuss tradeoffs between benefits and harms across options to be able to decide which option is most aligned with their informed values and preferences. Patient decision aids are tools that can support this process by providing information on options, benefits, and harms, and helping patients clarify their values for the outcomes of the options [62]. In a Cochrane review of 209 trials, patient decision aids were shown to be effective in increasing patient knowledge, supporting realistic expectations, and minimising decisional conflict [62].

However, current TKA/PKA patient decision aids are not optimal for adults with knee osteoarthritis considering these surgeries because they are outdated, based on low-quality evidence (e.g. non-randomised controlled trials (non-RCTs)), or not publicly available [63]. High-quality evidence is therefore needed on the effectiveness of TKA/PKA compared to other treatment options to inform healthcare professionals and patients during the decision-making process [63, 64].

A recent Cochrane review assessed the benefits and harms of surgical intervention for the management of mild to moderate knee osteoarthritis [65]. The authors acknowledged a priori that TKA and PKA were not indicated for milder knee osteoarthritis, but wanted to demonstrate the evidence. Their review found no RCTs on the effectiveness of TKA and PKA for adults with mild to moderate knee osteoarthritis [65]. Our review differs from [65], as we are interested in TKA/PKA and non-surgical interventions for the knee for moderate to severe osteoarthritis.

This review was conducted according to the guidelines recommended by the Cochrane Musculoskeletal Editorial Board [66].

OBJECTIVES

To assess the benefits and harms of TKA and PKA for people with moderate to severe knee osteoarthritis compared to placebo, sham (efficacy), or non-surgical interventions for the knee (effectiveness).

METHODS

We followed the Methodological Expectations of Cochrane Intervention Reviews (MECIR) when conducting the review and PRISMA 2020 for the reporting [67, 68]. Two amendments were made to the protocol's methods. First, the minimal clinically important differences (MCIDs) were updated to use outcome measure specific values, enhancing the clinical relevance and accuracy of result interpretation. Second, the methods for normalising serious adverse events were specified.

Criteria for considering studies for this review

Types of studies

We only included RCTs. All other study designs were excluded. There were no language restrictions for eligible trials that met the following eligibility criteria.

Types of participants

Adults (aged 18 years or older) with a diagnosis of moderate to severe osteoarthritis in one or both knees and eligible for surgical treatment. We excluded participants with other joint conditions (e.g. rheumatoid arthritis) or having TKA/PKA because of trauma (e.g. fracture), infection, or revision. We included trials with a mixture of participants with different joint conditions only if we were able to extract data on the eligible participants.

Types of interventions

We included TKA or PKA interventions using any surgical approach compared to placebo, sham, or any non-surgical interventions for the knee including physiotherapy, weight loss, pharmacotherapy, braces, insoles, and any intraarticular injection (e.g. stem cells, glucocorticoids, platelet-rich plasma).

For this review, we defined placebo surgery as "a procedure in which there is routine delivery of most of the operation, but the presumed active components, or the essential or critical surgical element, are removed" [69].

We defined sham surgery as entirely simulated surgery or small superficial incisions only, without breaking the skin or with very minimal incision [70].

Eligible studies could include usual care non-surgical interventions that may also have been provided as a co-intervention in the TKA or PKA group. Non-surgical interventions include rehabilitation programmes, weight loss, braces, insoles, pharmacotherapy, or any intraarticular injection (e.g. stem cells, glucocorticoids, platelet-rich plasma), while co-interventions could include, but were not limited to, pharmacotherapy, alternative therapies, preoperative rehabilitation programmes, and lifestyle changes.

We excluded patellofemoral arthroplasties and TKA or PKA revisions. We also excluded studies comparing two or more different types of surgery (e.g. different prosthesis, partial versus total, and surgical approaches).

Outcome measures

Major outcomes

Based on the OMERACT-OARSI core domain set for measurement in clinical trials of hip and knee osteoarthritis [71], and on the proposed outcomes of Cochrane Musculoskeletal, the major outcomes of interest were the following.

1. Pain, with a hierarchy of 11 levels (when more than one is reported, we used the highest on the list):
 - a. Pain overall
 - b. Pain on walking
 - c. Western Ontario and McMaster Universities Arthritis (WOMAC) index pain subscale
 - d. Pain on activities other than walking
 - e. WOMAC global scale
 - f. Lequesne osteoarthritis index global score
 - g. Other algofunctional scales (e.g. Knee Injury and Osteoarthritis Outcome Score (KOOS))
 - h. Patient's global assessment
 - i. Physician's global assessment
 - j. Other continuous pain outcomes
 - k. Non-continuous pain outcome (e.g. binary)
2. Physical function, with a hierarchy of eight levels (when more than one is reported, we used the highest on the list):
 - a. Global disability score
 - b. Walking disability
 - c. WOMAC disability subscore
 - d. Composite disability scores other than WOMAC
 - e. Knee-specific functional scales (e.g. KOOS)
 - f. Disability other than walking
 - g. WOMAC global scale
 - h. Lequesne osteoarthritis index global score
 - i. Other algofunctional scale
3. Knee surgery (including revision or subsequent operation for the TKA/PKA group or initial TKA/PKA for the non-surgical group)

4. Patient satisfaction with treatment outcomes (e.g. satisfaction with postoperative function)
5. Health-related quality of life
6. Serious adverse events (as defined by study authors, likely infection, deep vein thrombosis, pulmonary embolism, blood transfusion, cardiac events, death)
7. Withdrawals due to adverse events

Minor outcomes

1. Performance-based tests (e.g. Timed Up and Go (TUG), 6-Minute Walk Test)
2. All adverse events (e.g. swelling)
3. Cost and cost-effectiveness

Timing of outcome assessments

We measured follow-up outcomes at short term (≤ 6 months), intermediate term (6 to 12 months), and long term (≥ 1 year). For efficacy outcomes, the primary analysis was at long-term follow-up. For adverse events, we reported and analysed the events at all time points.

Minimal clinically important differences

We used the following MCIDs for interpreting our results.

1. Visual analogue scale (VAS) pain (0 to 100): 22.6 [72]
2. KOOS pain (0 to 100): 15 [73]
3. KOOS quality of life (0 to 100): 15.6 [73]
4. KOOS activity of daily living (minimal detectable change) (0 to 100): 17.4 [74]
5. EQ-5D Descriptive Index (0 to 100): 0.3 [75]

Search methods for identification of studies

Electronic searches

We searched the following databases:

1. the Cochrane Central Register of Controlled Trials (CENTRAL; 2025, Issue 1) via Ovid (searched 27 January 2025);
2. MEDLINE via Ovid (1946 to 27 January 2025);
3. Embase via Ovid (1946 to 27 January 2025).

We also searched the following trial registries:

1. ClinicalTrials.gov (www.clinicaltrials.gov) (searched 27 January 2025);
2. World Health Organization International Clinical Trials Registry Platform (WHO ICTRP) (www.who.int/ictcp/en/) (searched 27 January 2025).

We searched all databases from 2010 to the present because of major practice changes in TKA surgical techniques [76]. Additionally, the Cochrane review of surgical interventions for mild to moderate knee osteoarthritis, which searched databases from inception to 2018, did not find any trials of TKA or PKA [65].

See [Supplementary material 1](#) for the search strategies.

Searching other resources

We checked the reference lists of all primary trials and reviewed articles for additional references. We searched for errata or retractions from included trials published in full text on PubMed (www.ncbi.nlm.nih.gov/pubmed) and the Retraction Watch Database (retractiondatabase.org) for the first author of trials, and we reported the date this was done within the review. We tracked citations of included trials using the Bond University Systematic Review Accelerator (sr-accelerator.com/#/).

Data collection and analysis

Selection of studies

We excluded duplicates. Four reviewers (LPB, DS, SA, NN) working in pairs independently screened titles and abstracts against eligibility criteria. We retrieved the full-text study reports/publication of potentially relevant trials (eligible or potentially eligible/unclear), and two review authors (LPB, SA) independently screened the full text for inclusion in the review, and recorded the reasons for exclusion of excluded studies. Any disagreements were resolved through discussion with all review authors. We collated multiple reports of the same study so that each study, rather than each report, was the unit of interest in the review. We recorded the selection process in sufficient detail to complete a PRISMA flow diagram (Figure 1) [77, 78] and 'Characteristics of excluded studies' table (Supplementary material 3).

Figure 1. Study flow diagram.

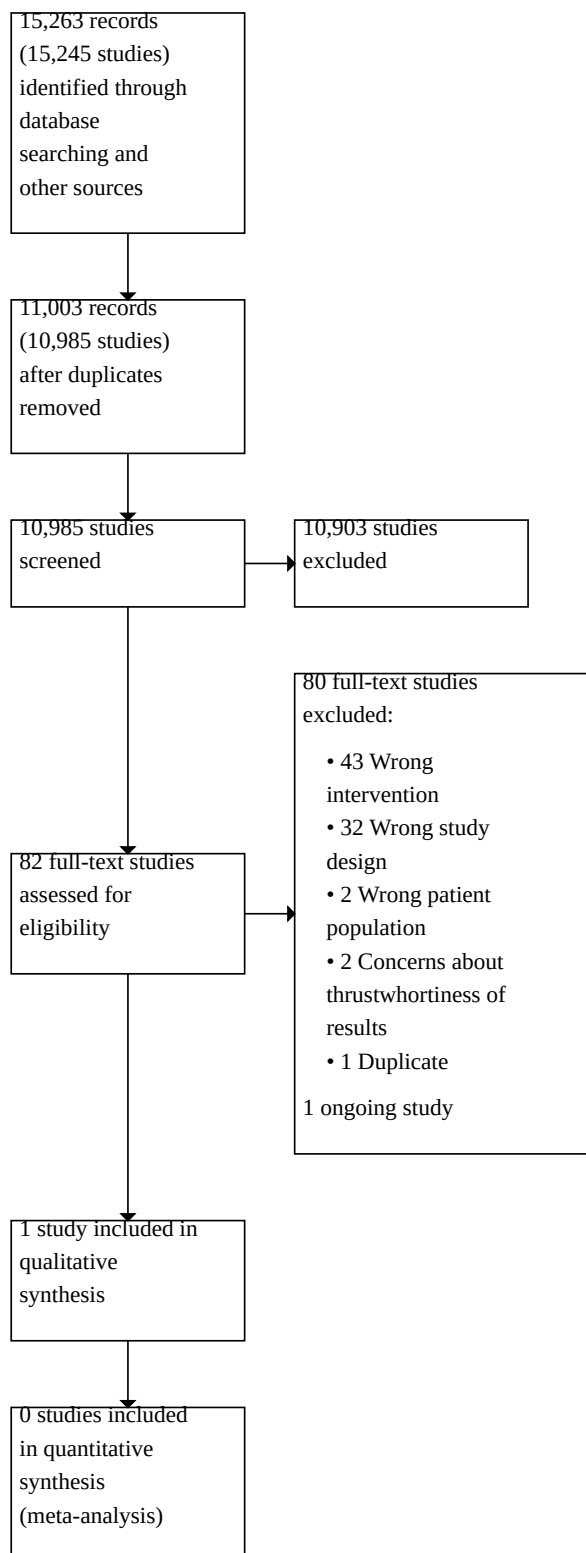


Figure 1. (Continued)

Data extraction and management

Two review authors (LPB, SA) independently extracted data from included trials using a data collection form. We extracted data on the following characteristics.

1. Study methods: study design and characteristics, total duration of study, details of any 'run-in' period, number of study centres and location, study setting, withdrawals, and date of study.
2. Participants: N, mean age, age range, sex, disease duration, severity of condition, comorbidities, sociodemographics, ethnicity, diagnostic criteria, important baseline data, inclusion criteria, and exclusion criteria.
3. Interventions: total number of intervention arms, intervention details, comparison details, co-intervention details.
4. Outcomes: all major and minor outcomes specified and collected, and time points reported. We used PlotDigitizer to extract data from graphs or figures [79].
5. Other: funding for study, and declarations of interest of study authors.

We extracted the number of events and number of participants per treatment group for dichotomous outcomes, and means and standard deviations (SDs) (or statistics from which SDs could be derived) and number of participants per treatment group for continuous outcomes. We noted in the 'Outcomes Reported by Studies' table (Supplementary material 8) whether outcome data were not reported in a usable way and when data were transformed or estimated from a graph. Any disagreements were resolved by consensus within the review author group. We transferred data from the data collection form to RevMan [80], double-checking that data had been entered correctly by comparing the data presented in the systematic review with the study reports.

If more than one measure for an outcome was reported, they were prioritised based on the hierarchy of major outcomes. If data for more than one time point was provided, we planned to use the longest follow-up for each time point (e.g. closest to six months for short term) for the meta-analysis.

If both unadjusted and adjusted values for the same outcome were reported, we extracted unadjusted values. If data were analysed based on an intention-to-treat (ITT) sample and another sample (e.g. per-protocol, as-treated), ITT data were extracted. For adverse effects, as-treated data were extracted. If continuous outcome measures were reported in multiple ways, we followed the hierarchy below.

1. Final values adjusted for baseline characteristics
2. Final values
3. Change from baseline

Risk of bias assessment in included studies

Two review authors (LPB, SA) independently assessed the risk of bias for each study using the RoB 1 tool as outlined in the *Cochrane Handbook for Systematic Reviews of Interventions* [81].

Any disagreements were resolved by discussion or within the review author group. We assessed risk of bias according to the following domains.

1. Random sequence generation
2. Allocation concealment
3. Blinding of participants and personnel
4. Blinding of outcome assessment
5. Incomplete outcome data
6. Selective outcome reporting
7. Other bias (e.g. potential sources of bias reported by authors, participant selection, study design)

If we found cluster trials, we planned to follow appropriate methods in assessing risk of bias as outlined in the *Cochrane Handbook for Systematic Reviews of Interventions* [82].

We rated each domain as low, high, or unclear risk of bias, and provided a quote from the study report together with a justification for our judgement in the risk of bias table (Supplementary material 2). We summarised the risk of bias judgements across different trials for each of the domains listed. Given the nature of the trials, we considered bias due to lack of blinding for all outcomes. We also considered the impact of missing data by key outcomes.

Where information on risk of bias related to unpublished data or correspondence with a trialist, we noted this in the risk of bias table (Supplementary material 2).

When considering treatment effects, we planned to account for the risk of bias in the trial that contributed to that outcome.

We presented the figures generated by the RoB 1 tool to provide summary assessments of the risk of bias.

Measures of treatment effect

We were not able to conduct a meta-analysis given there was only one included study. Hence, we reported the treatment effect as was reported in this eligible study. In the case of sufficient similar data, our plans for meta-analysis were as follows.

We planned to analyse dichotomous data as risk ratios (RRs) or Peto odds ratios (ORs) when the outcome was a rare event (approximately less than 10%), and use 95% confidence intervals (CIs). We planned to analyse continuous data as mean difference (MD) or standardised mean difference (SMD), depending on whether the same scale was used to measure the outcome, and 95% CIs. We planned to enter data presented as a scale with a consistent direction of effect across trials.

If different scales were used to measure the same conceptual outcome (e.g. disability), we planned to instead calculate SMDs, with corresponding 95% CIs. We would have back-translated SMDs to a typical scale (e.g. 0 to 10 for pain) by multiplying the SMD by a typical among-person SD (e.g. the SD of the control group at baseline from the most representative study) [82, 83].

For dichotomous outcomes, we planned to calculate the number needed to treat for an additional beneficial outcome (NNTB) or the number needed to treat for an additional harmful outcome (NNTH) from the control group event rate and the risk ratio using the Visual Rx NNT calculator [84]. We would have calculated NNTB or NNTH for continuous measures using the Wells calculator (available at the CMSG Editorial office, musculoskeletal.cochrane.org/). We would have used the MCID to calculate NNTB or NNTH, assuming a MCID of 1.5 points on a 10-point scale for pain, and 10 points on a 100-point scale for function or disability for input into the calculator [85].

For dichotomous outcomes, we planned to calculate the absolute per cent change from the difference in the risks between the intervention and control group using GRADEpro GDT [86], expressed as a percentage. We would have calculated the relative per cent change as the absolute value of risk ratio – 1, expressed as a percentage.

For continuous outcomes, we planned to calculate the absolute per cent change by dividing the mean difference by the scale of the measure, expressed as a percentage. We would have calculated the relative difference as the absolute benefit (mean difference) divided by the baseline mean of the control group, expressed as a percentage.

In the case of sufficient trials, we planned to provide the absolute per cent change, the relative per cent change from baseline, and the NNTB or NNTH (NNTB or NNTH is provided only when the outcome shows a clinically significant difference) in the 'Synthesis of results' section and the 'What happens' column of the summary of findings table (Summary of findings 1).

Unit of analysis issues

The unit of analysis was at the individual level. If we included trials randomising knees to different groups, we planned to conduct sensitivity analysis and consider potential sources of other bias. We did not find RCTs with non-standard designs (e.g. cross-over, multiple treatment arms). If multiple study arms were reported in a single study, we planned to include only the relevant arms. We planned to pool arms together when possible. If two comparisons (e.g. physiotherapy versus TKA and weight loss versus TKA) were combined in the same meta-analysis, we planned to halve the control group to avoid double-counting. For continuous outcomes, we planned to halve the control group and the mean and SD remained the same. For binary outcomes, we planned to halve the number of events in the numerator and denominator in the control group.

To normalise data included from the trial, serious adverse events were converted to the number of events per 100 participants per one year.

Dealing with missing data

We contacted investigators to obtain missing numerical outcome data where possible (e.g. when a study was identified as abstract only, or when data were not available for all participants). Where this was not possible, and the missing data were thought to introduce serious bias, we planned to explore the impact of including such trials in the overall assessment of results by sensitivity analysis. We planned to clearly describe any assumptions and imputations to handle missing data and to explore the effect of imputation by sensitivity analyses.

For dichotomous outcomes (e.g. number of withdrawals due to adverse events), we planned to calculate the withdrawal rate using the number of participants randomised in the group as the denominator.

For continuous outcomes (e.g. mean change in pain score), we planned to calculate the MD or SMD based on the number of participants analysed at that time point. If the number of participants analysed was not presented for each time point, we would use the number of randomised participants in each group at baseline.

If SDs were missing, we planned to compute them from other statistics such as standard errors, CIs or P values, according to the recommendations in the *Cochrane Handbook for Systematic Reviews of Interventions* [87]. If SDs could not be calculated, we planned to impute (e.g. from other trials in the meta-analysis).

Reporting bias assessment

We did not conduct a funnel plot to explore possible small-study biases given that we did not include more than 10 trials [88]. In interpreting funnel plots, we planned to examine the different possible reasons for funnel plot asymmetry as outlined in Chapter 13 of the *Cochrane Handbook for Systematic Reviews of Interventions* and relate this to the results of the review. If we were able to pool more than 10 trials, we would have undertaken formal statistical tests to investigate funnel plot asymmetry, following the recommendations in Section 13.3 of the *Cochrane Handbook for Systematic Reviews of Interventions* [68].

To assess outcome reporting bias, we checked study protocols against published reports. For trials published after 1 July 2005, we screened the WHO ICTRP (www.who.int/ictip/en/) for the a priori study protocol. We evaluated whether selective reporting of outcomes was present.

Synthesis methods

Main planned comparisons

We were not able to conduct the a priori planned meta-analysis because there was only one eligible study. We did not stratify TKA and PKA because there were no eligible PKA trials.

The plan was to undertake meta-analyses for the following comparisons using RevMan [80].

1. Comparison 1: TKA and PKA versus sham/placebo
2. Comparison 2: TKA and PKA versus usual care non-surgical interventions, including, but not limited to, rehabilitation programmes, weight loss, braces, insoles, pharmacotherapy, any intraarticular injection (e.g. stem cells, glucocorticoids, platelet-rich plasma), when usual care is not a co-intervention of TKA
3. Comparison 3: TKA and PKA plus usual care non-surgical interventions versus usual care non-surgical interventions alone

We planned to use a random-effects model and perform a sensitivity analysis with the fixed-effect model. The primary analysis would have included all trials regardless of risk of bias.

Given that meta-analysis was precluded by insufficient similar data, we synthesised our results descriptively using different methods

as outlined in Chapter 12 of the *Cochrane Handbook for Systematic Reviews of Interventions* [89].

We planned to assess clinical and methodological diversity in terms of participants, interventions, outcomes, and study characteristics for the included trials to determine whether meta-analysis was appropriate. We planned to do this by observing these data from the data extraction tables. We planned to assess statistical heterogeneity by visual inspection of the forest plot to look for obvious differences in results between the trials, and using the I^2 and χ^2 statistics.

As recommended in the *Cochrane Handbook for Systematic Reviews of Interventions* [87], we interpreted the I^2 value as follows:

1. 0% to 40%: might not be important;
2. 30% to 60%: may represent moderate heterogeneity;
3. 50% to 90%: may represent substantial heterogeneity; and
4. 75% to 100%: represents considerable heterogeneity.

As noted in the *Cochrane Handbook for Systematic Reviews of Interventions* [87], we planned to keep in mind that the importance of I^2 depends on: (i) the magnitude and direction of effects; and (ii) the strength of evidence for heterogeneity.

We planned to interpret the χ^2 test as $P \leq 0.10$ indicating evidence of statistical heterogeneity.

Had we identified substantial heterogeneity, we would have reported it and investigated possible causes following the recommendations in Section 10.10 of the *Cochrane Handbook for Systematic Reviews of Interventions* [87].

Investigation of heterogeneity and subgroup analysis

Insufficient data precluded subgroup analyses. Had there been sufficient data, we would have carried out the following analyses.

1. Age: < 55 years, 55 to 75 years, > 75 years
2. Sex/gender
3. Ethnicity – when possible, white, Aboriginal, South Asian, Chinese, Black, Latin American, Arab, Southeast Asian, West Asian, multiple ethnicities, other ethnicities; or authors reported grouping [90]
4. Non-surgical intervention types (e.g. education, physiotherapy, pharmacotherapy)
5. Implant type (e.g. fixed, mobile)
6. Placebo/sham

We chose these subgroups because: 1) knee osteoarthritis is more prevalent in white women [91]; 2) there are disparities in demand and recommendation of TKA/PKA based on age, sex/gender, and ethnicity [92, 93, 94]; 3) there are differences in preoperative expectations and postoperative outcomes based on sex and ethnicity [95, 96, 97]; 4) there are different types of non-surgical interventions for knee osteoarthritis (e.g. education, pharmacotherapy) with a different effect on management [15]; 5) implant types can potentially influence outcomes [98]; 6) placebo and sham surgery are not the same concept [70], and it is unclear whether this would make a difference in this setting.

We planned to use major outcomes in subgroup analyses if sufficient data were available.

We planned to use the formal test for subgroup interactions in RevMan [80], and to use caution in the interpretation of subgroup analyses as advised in Chapter 10 of the *Cochrane Handbook for Systematic Reviews of Interventions* [87]. The magnitude of the effects would have been compared between the subgroups by means of assessing the overlap of the CIs of the summary estimate. Non-overlap of the CIs would indicate statistical significance.

Equity-related assessment

We did not plan for an equity-related assessment in our protocol. As part of our data analysis, we examined the composition of the populations of the included study to judge the applicability of the results in different populations. The assessment was incorporated into the [Discussion](#).

Sensitivity analysis

We were not able to carry out the a priori planned sensitivity analyses because there was only one eligible study.

1. Unclear or high risk of selection bias (random sequence generation and allocation concealment)
2. Unclear or high risk of detection bias (blinding of outcome assessment)
3. Unclear or high risk of attrition bias (incomplete outcome data)

Certainty of the evidence assessment

We followed the guidelines in Chapter 15 of the *Cochrane Handbook for Systematic Reviews of Interventions* for interpreting results, and were aware of distinguishing a lack of evidence of effect from a lack of effect [99]. We based our conclusions only on findings from the narrative synthesis of the single trial included in this review. We avoided making recommendations for practice. Our implications for research suggested priorities for future research and outlined the remaining uncertainties in the area.

We created a summary of findings table ([Summary of findings 1](#)) comparing TKA to non-surgical treatments for the major outcomes of pain, physical function, knee surgery, patient satisfaction, health-related quality of life, serious adverse events, and withdrawals due to adverse events. We had planned to create a first summary of findings table comparing TKA or PKA versus placebo and a second summary of findings comparing TKA or PKA versus other interventions.

Two review authors (LPB, SA) independently assessed the certainty of the evidence, with any disagreements resolved by discussion or involving a third reviewer (JPP, NN, DS, or GW). We used the five GRADE considerations (study limitations, consistency of effect, imprecision, indirectness, and publication bias) to assess the certainty of the evidence for the prespecified outcomes as high, moderate, low, or very low. We justified, documented, and incorporated these judgements into the reporting of results for each outcome. We used the methods and recommendations described in Chapter 14 of the *Cochrane Handbook for Systematic Reviews of Interventions* [100].

We used GRADEpro GDT software to prepare [86] and version 3 of the GRADEpro view to display our summary of findings table. We justified all decisions to downgrade the certainty of evidence using footnotes and made comments to aid the reader's understanding of the review where necessary. We planned to provide the NNTB

or NNTH, absolute and relative per cent change in the 'What happens' column of the summary of findings table as described in the [Measures of treatment effect](#) section, except for the absolute difference for dichotomous outcomes, which is displayed by default in version 3 of the GRADEpro view.

Consumer involvement

No patients or health consumers were involved in this review.

RESULTS

Description of studies

Results of the search

The search identified 15,263 citations in databases and other sources ([Figure 1](#)). After the removal of 4260 duplicates, 11,003 titles and abstracts were screened, and 82 full texts were assessed for eligibility. We excluded 80 studies, identified one study as ongoing (Rognsvåg 2024 [[101](#), [102](#)], [Supplementary material 4](#)), and included one study reported across seven peer-reviewed publications in the review (Skou 2015 [[103](#), [104](#), [105](#), [106](#), [107](#), [108](#), [109](#)], [Supplementary material 2](#)).

Included studies

A description of the included study is presented in the 'Overview of Syntheses and Included Studies' table ([Table 1](#)) and in [Supplementary material 2](#). We requested additional outcome details from the study authors. Specific KOOS data at different time points during the first year were provided for each treatment group but were not included in the review. No further data on participants with moderate to severe osteoarthritis or adverse events were received.

We included one study conducted in Denmark (Skou 2015), a two-arm parallel-group RCT including 100 adults with primary mild to severe knee osteoarthritis ($n = 12$ with Kellgren-Lawrence score ≥ 2 , $n = 42$ with Kellgren-Lawrence score 3, $n = 46$ with Kellgren-Lawrence score 4). The study evaluated TKA followed by a 12-week non-surgical programme compared to the 12-week non-surgical programme alone (exercises, osteoarthritis education, dietary advice, insoles, pain medication) (see [Supplementary material 7](#)

for detailed participant characteristics). Eligibility criteria used by orthopaedic surgeons to assess TKA eligibility were not reported.

In the TKA followed by a non-surgical programme, 32 were females (64%), mean age was 65.8 years (SD 8.7), mean weight was 89.4 kg (SD 21.9), and mean body mass index (BMI) (kg/m^2) was 32.3 (SD 6.2). In the non-surgical programme, 30 were females (60%), mean age was 66.4 years (SD 8.7), mean weight was 91.1 kg (SD 16.1), and mean BMI (kg/m^2) was 32.0 (SD 5.8) (see [Supplementary material 7](#) for detailed participant characteristics).

Ongoing study

We identified one ongoing study (Rognsvåg 2024) ([Supplementary material 4](#)). Rognsvåg 2024 is a three-arm feasibility RCT including a total of 15 adults (five per study arm) comparing 1) a 12-week exercise programme, education, and internet-delivered cognitive behavioural therapy (iCBT), 2) TKA followed by a 12-week exercise programme, education, and iCBT, and 3) TKA followed by standard postoperative care. This feasibility study did not evaluate group differences or effect sizes since the objective was to assess feasibility.

Excluded studies

We excluded a total of 80 studies. Justifications for exclusion are presented in the 'Characteristics of excluded studies' table ([Supplementary material 3](#)). We excluded two studies because their study population consisted of adults with osteoarthritis secondary to osteonecrosis and due to concerns about the trustworthiness of their results, as determined by the Trustworthiness Screening Tool [[110](#)]. Since the author did not respond to requests for clarification regarding the identified issues, the decision was made to exclude these studies.

Risk of bias in included studies

We judged the study by Skou 2015 to be at low risk for selection (random sequence generation, allocation concealment), performance (blinding of participants and personnel), detection (blinding of outcome assessment), attrition (incomplete outcome data), reporting (selective reporting), and other biases ([Figure 2](#); [Figure 3](#)).

Figure 2. Risk of bias graph summarising the review authors' judgements across key domains for each included study.

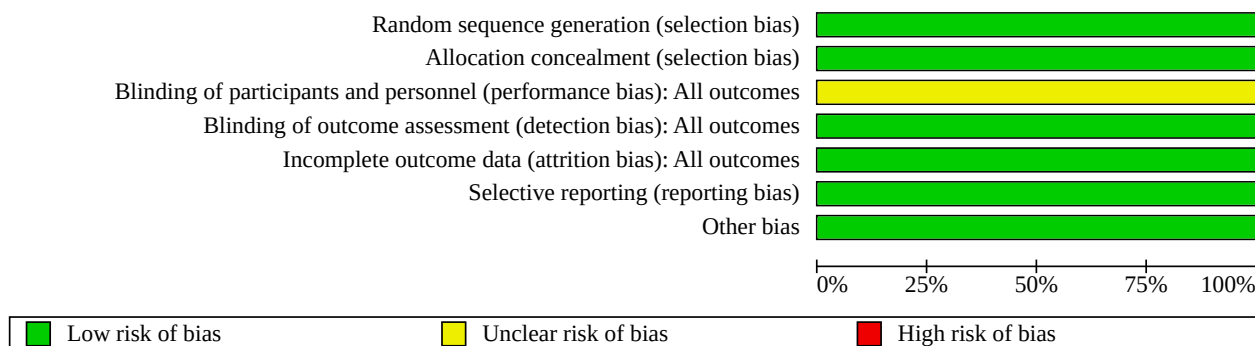
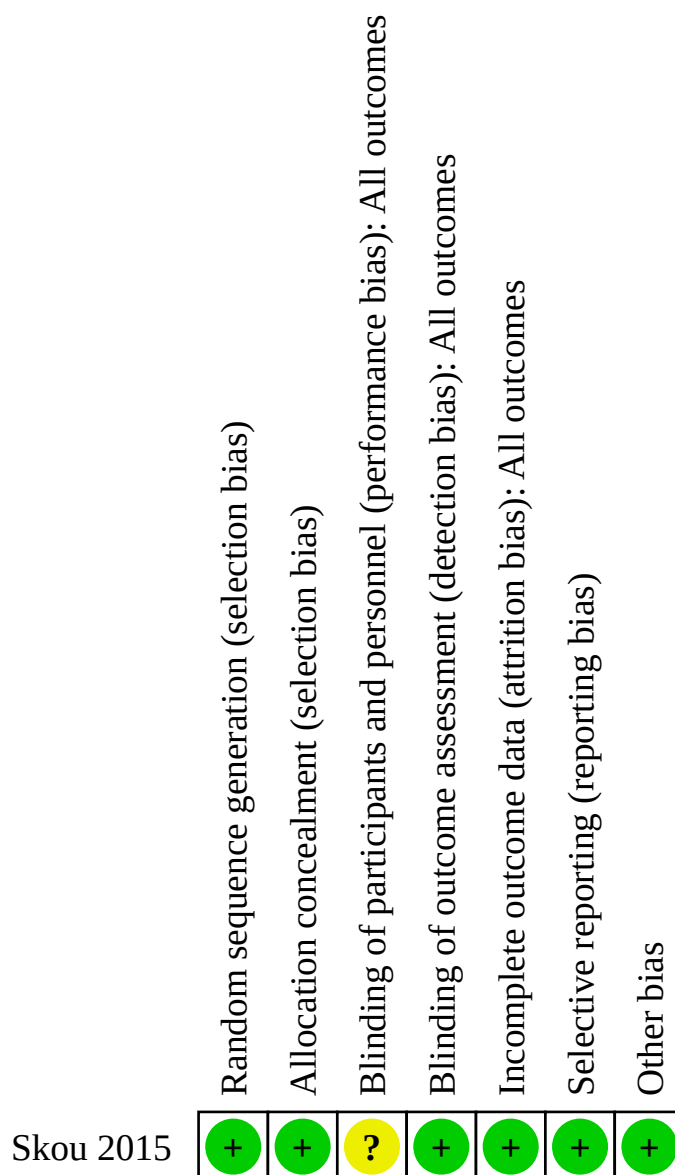


Figure 3. Risk of bias summary of the review authors' assessments of each included study across multiple domains.



Allocation

We judged Skou 2015 to have a low risk of selection bias. The study provided sufficient details on the methods of randomisation and allocation concealment.

Blinding

We judged Skou 2015 to have an unclear risk of performance bias. Due to the nature of the intervention, blinding of participants and personnel was not feasible, which could have influenced

outcome measures subject to judgement, potentially introducing performance bias.

We judged Skou 2015 to have a low risk of detection bias. Sufficient details were provided indicating that outcome assessors were blinded to treatment allocation, minimising the potential for detection bias.

Incomplete outcome data

We judged Skou 2015 to have a low risk of attrition bias. The study included a detailed participant flow diagram, provided reasons for missing follow-up data, reported similar rates of loss to follow-up between groups, and did not have any withdrawals.

Selective reporting

We judged Skou 2015 to have a low risk of reporting bias. The trial was registered; the protocol was published a priori; and all primary and secondary outcomes were reported. However, the study included outcomes such as length of hospital stay and number of treatments or consultations with other practitioners that were not identified in the trial registry or published protocol.

Other potential sources of bias

We judged Skou 2015 to have a low risk of other biases, with no additional concerns identified.

Synthesis of results

The main comparison is presented in [Summary of findings 1](#).

TKA combined with a non-surgical programme compared to a non-surgical programme alone

Skou 2015 evaluated adults who were randomised to TKA followed by a 12-week non-surgical programme compared to the same 12-week non-surgical programme alone (see [Supplementary material 5](#) for analyses).

Major outcomes

All major outcomes were assessed by Skou 2015 except for patient satisfaction with outcomes (see detailed description of outcome assessment in [Supplementary material 8](#)).

Pain

Unilateral TKA may reduce pain at a level that is clinically important. Compared to a non-surgical programme alone, adults with mild to severe knee osteoarthritis who had TKA followed by a non-surgical programme had improved pain for at one year (KOOS pain MD 17.60, 95% CI 8.25 to 26.95; 1 study, 100 participants; low-certainty evidence, Analysis 1.1), and this level of improvement met the MCID. The certainty of evidence was low given the imprecision of results included (did not achieve the optimal information size) and risk of bias (participants were not blinded).

Compared to a non-surgical programme alone, adults with mild to severe knee osteoarthritis who had TKA followed by a non-surgical programme had no difference in improvement in pain at three months (VAS MD 7.90, 95% CI -5.48 to 21.28; 1 study, 100 participants, Analysis 1.11). Compared to a non-surgical programme alone, adults with mild to severe knee osteoarthritis who had TKA followed by a non-surgical programme had improvement in pain at two years (KOOS pain MD 17.30, 95% CI 9.20 to 25.40; 1 study, 100 participants, Analysis 1.7).

Physical function

Unilateral TKA may improve physical function, but the improvement might not be clinically important. Compared to a non-surgical programme alone, adults with mild to severe knee osteoarthritis who had TKA followed by a non-surgical programme

had improved physical function at one year (KOOS activities of daily living (ADL) MD 12.40, 95% CI 3.06 to 21.74; 1 study, 100 participants; low-certainty evidence, Analysis 1.2), but the improvement did not meet the MCID. The certainty of evidence was low given the imprecision of results included (did not achieve the optimal information size) and risk of bias (participants were not blinded).

Compared to a non-surgical programme alone, adults with mild to severe knee osteoarthritis who had TKA followed by a non-surgical programme improved physical function at two years (KOOS ADL MD 15.10, 95% CI 7.69 to 22.51; 1 study, 100 participants, Analysis 1.8).

Knee surgery

Unilateral TKA may reduce the need for a follow-up knee surgery (revision or subsequent knee surgery and initial knee surgery). Compared to a non-surgical programme alone, adults with mild to severe knee osteoarthritis who had TKA followed by a non-surgical programme had reduced need for follow-up knee surgery (revision or subsequent operation for the TKA group or initial TKA for the non-surgical group). At one year, 0 adults (of $n = 49$) receiving TKA followed by a non-surgical programme underwent a revision or subsequent operation, and 13 adults (of $n = 50$) receiving the non-surgical programme underwent a TKA (RR 0.04, 95% CI 0.00 to 0.62; 1 study, 99 participants; low-certainty evidence, Analysis 1.3). The certainty of evidence was low given the imprecision of results included (did not achieve the optimal information size) and risk of bias (participants were not blinded).

At two years, only data on TKA and revision surgery were available: four adults (of $n = 49$) receiving TKA followed by a non-surgical programme underwent a revision or subsequent operation, and 17 adults receiving the non-surgical programme (of $n = 50$) underwent a TKA or another knee surgery (RR 0.24, 95% CI 0.09 to 0.66; 1 study, 99 participants, Analysis 1.10).

Patient satisfaction with treatment outcomes

The included trial did not assess patient satisfaction with treatment outcomes.

Health-related quality of life

There is probably no clinically important difference in health-related quality of life between unilateral TKA followed by a non-surgical programme and non-surgical programme alone at one year (EQ-5D Descriptive Index MD 0.09, 95% CI 0.01 to 0.18; 1 study, 100 participants; moderate-certainty evidence, Analysis 1.4). The certainty of evidence was moderate given the risk of bias (participants were not blinded). The certainty of evidence was not downgraded for imprecision because the effect estimates and the CI did not cross the line of the MCID.

Compared to a non-surgical programme alone, adults with mild to severe knee osteoarthritis who had TKA followed by a non-surgical programme did not have improved quality of life at two years (KOOS quality of life MD 24.10, 95% CI 15.75 to 32.45; 1 study, 100 participants, Analysis 1.9).

Serious adverse events

Unilateral TKA may increase serious adverse events, but the evidence is very uncertain. At one year, there were 48 serious adverse events per 100 adults per one year in the TKA followed by a non-surgical programme group (e.g. deep venous thrombosis, deep infection, supra-condylar femur fracture) and 12 serious

adverse events per 100 adults per one year in the non-surgical programme alone group (e.g. stiffness leading to manipulation under anaesthesia, cerebral thrombosis) (RR 4.00, 95% CI 1.79 to 8.94; 1 study, 100 participants; very low-certainty evidence, Analysis 1.5). The certainty of evidence was very low given the imprecision of results included (did not achieve the optimal information size), indirectness (the definition of serious adverse events being too inclusive), and risk of bias (participants were not blinded).

Withdrawals due to adverse events

There may be no difference between unilateral TKA followed by a non-surgical programme and non-surgical programme alone, but the evidence is very uncertain. Compared to a non-surgical programme alone, adults with mild to severe knee osteoarthritis who had TKA followed by a non-surgical programme did not differ in withdrawals due to adverse events ($n = 0$ for TKA followed by a non-surgical programme, $n = 0$ for non-surgical programme alone, Not estimable; 1 study, 100 participants; very low-certainty evidence, Analysis 1.6). The certainty of evidence was very low given the imprecision of results included (low number of events) and risk of bias (participants were not blinded).

Minor outcomes

Performance-based tests

Compared to a non-surgical programme alone, adults with mild to severe knee osteoarthritis who had TKA followed by a non-surgical programme had better mobility at one year (TUG MD 1.20, 95% CI 0.46 to 1.94; 1 study, 100 participants, Analysis 2.1) and two years (TUG MD 1.50, 95% CI 0.71 to 2.29; 1 study, 100 participants, Analysis 2.2).

All adverse events

Compared to a non-surgical programme alone, adults with mild to severe knee osteoarthritis who had TKA followed by a non-surgical programme reported an increased number of adverse events. At one year, there were 87 non-serious adverse events reported in the TKA followed by a non-surgical programme group and 67 non-serious adverse events in the non-surgical programme alone group.

Cost and cost-effectiveness

Compared to a non-surgical programme alone, TKA followed by a non-surgical programme is more costly at one year (TKA followed by a non-surgical programme mean EUR 16,343; non-surgical programme alone mean EUR 7028) and two years (TKA followed by a non-surgical programme mean EUR 6733; non-surgical programme alone mean EUR 7486).

Compared to a non-surgical programme alone, TKA followed by a non-surgical programme was more cost-effective at one year (TKA followed by a non-surgical programme mean (SD) 0.858 (0.195); non-surgical programme alone mean (SD) 0.795 (0.153), MD 0.06, 95% CI -0.00 to 0.13; 1 study, 100 participants, Analysis 2.3) and two years (TKA followed by a non-surgical programme mean (SD) 0.878 (0.155); non-surgical programme alone mean (SD) 0.758 (0.199), MD 0.12, 95% CI 0.05 to 0.19; 1 study, 100 participants, Analysis 2.4).

Compared to a non-surgical programme alone, TKA followed by a non-surgical programme is not more cost-effective at two years (32,611 Euros/quality-adjusted life years (QALYs), cost-effectiveness probability of 23.2%).

DISCUSSION

Summary of main results

This review assessed the benefits and harms of TKA compared to non-surgical interventions for the knee for people with mild to severe knee osteoarthritis and included one study comparing TKA followed by a 12-week non-surgical programme to the same non-surgical programme alone. Based on the GRADE assessment ([Summary of findings 1](#)), we found low-certainty evidence that TKA followed by a non-surgical programme may improve pain at a level that is clinically important; may improve physical function at a level that is not clinically important; and may reduce the need for a follow-up knee surgery (revision or subsequent knee surgery and initial knee surgery) compared to a non-surgical programme alone. The included study did not evaluate patient satisfaction with surgical (TKA/PKA) and non-surgical treatment outcomes. Moderate-certainty evidence shows there is likely no clinically important difference in health-related quality of life between TKA followed by a non-surgical programme and a non-surgical programme alone. TKA followed by a non-surgical programme may increase the risk of serious adverse events compared to a non-surgical programme alone, but the evidence is very uncertain. There may be no difference in withdrawals due to adverse events between TKA followed by a non-surgical programme and a non-surgical programme alone, but the evidence is very uncertain.

Limitations of the evidence included in the review

The quality and robustness of the evidence in this review is limited by the number of included studies. Since only one study met the eligibility criteria, this review only partially addresses its objective and provides evidence on the benefits and harms of TKA compared to non-surgical intervention for the knee. The included study investigated all outcomes of interest except patient satisfaction with outcomes. The certainty of evidence was very low to moderate due to imprecision and the inherent limitations of a single study.

We judged Skou 2015 to be at low risk for selection bias, performance bias, detection bias, attrition bias, reporting bias, and other biases. Lack of blinding of participants and personnel introduces the potential for performance bias.

The generalisability of the findings is limited by the fact that this review relies on evidence from a single RCT. The study was conducted in Denmark between 2011 and 2013, where unique cultural, healthcare, and lifestyle factors may have influenced the results. Denmark's healthcare system provides universal coverage through tax-funded public healthcare, ensuring free access to most medical services, including hospital care and surgeries [111]. While physiotherapy is partially covered, requiring co-payments, the comprehensive insurance coverage for surgical interventions may impact treatment decisions and patient outcomes differently than in healthcare systems with greater financial barriers [112, 113]. Additionally, participants were predominantly female (62%) and had a mean age of 66 years, which aligns with the demographic typically affected by osteoarthritis worldwide [114]. Most participants were classified as obese, with an average BMI of 32.2 kg/m², indicating that the study largely represented a more unique patient demographic. While these characteristics make the findings relevant to certain groups, they may not be broadly applicable to more diverse osteoarthritis populations, such as younger adults, those with different BMI ranges, those from varying

ethnic and cultural backgrounds, or different healthcare systems. Regardless, the study offers insights into the effectiveness of TKA and non-surgical interventions within a publicly funded healthcare system.

While the study focused on individuals with moderate to severe osteoarthritis, 12% of adults had milder osteoarthritis despite the eligibility criteria specifying moderate to severe cases. This deviation raises concerns about the consistency of participant selection, specifically the TKA appropriateness criteria used by the orthopaedic surgeons when determining TKA eligibility, given the limited detail provided. Additionally, the study excluded individuals with a mean pain score above 60 mm on a 100-millimetre VAS in the previous week, resulting in the inclusion of only participants with milder pain. This exclusion criterion significantly limits the clinical applicability of the findings, as it does not reflect the broader population of patients with moderate to severe osteoarthritis who typically experience higher levels of pain and may be more likely to consider surgical intervention [115]. It is also unclear whether participants received an adequate trial of conservative treatment before surgery, as recommended by clinical practice guidelines for knee osteoarthritis [5, 14, 15]. The lack of detail on surgical eligibility and adequate trial of conservative treatment limits the ability to assess whether surgery was appropriate. To support informed quality decisions and follow evidence-based practices, patient decision aids should be used to facilitate shared decision-making [62]. Together, these factors may introduce variability in treatment response and outcomes, potentially influencing the study's conclusions and limiting their generalisability to clinical practice.

Although the study reported a high number of serious adverse events (48 events per 100 participants per year in the TKA group, 12 events per 100 participants per year in the non-surgical programme group), it is notable that none of these events led to participant withdrawal (no withdrawals due to adverse events were recorded). This raises concerns about how serious adverse events were defined and whether the classification was too inclusive. According to our [Methods](#) section, serious adverse events were reported based on study-specific definitions, but in this case, the definition may not align with more commonly accepted classifications of severe complications. In clinical practice, serious adverse events typically include venous thromboembolism, deep infection or sepsis, myocardial infarction, heart failure or cardiac arrest, periprosthetic fractures, stroke or cerebrovascular accident, and pneumonia [22]. However, it is possible that the study classified less clinically significant complications as serious adverse events, given that the study definition encompassed any event with the potential to compromise clinical outcomes, cause disability, require hospital care, or pose a life-threatening risk, regardless of whether the event was directly related to the intervention [22, Skou 2015]. A more nuanced classification system, such as the Clavien-Dindo system [116], which categorises complications based on their clinical impact and required interventions, may provide a better assessment of the findings (e.g. certainty of evidence). Insufficient detail in reporting of serious adverse events prevented further analysis of the data. Given the potentially broad definition used in this study, these results should be interpreted with caution since they may reflect an overly inclusive definition of serious adverse events rather than true clinical severity.

Limitations of the review processes

The strengths of this review include following standard Cochrane procedures [82] and using a comprehensive search strategy for all relevant major electronic databases and trial registries with no language restrictions. The search strategy covered studies published in the last 15 years (2010 to 2025), as surgical approaches have changed since before this period [76]. Therefore, even if only one study was identified, the risk of overlooking eligible studies is small, especially because the Cochrane review of surgical interventions for mild to moderate knee osteoarthritis, which searched databases from inception to 2018, did not find any trials of TKA or PKA [65]. Furthermore, efforts were made to retrieve all relevant data, including contacting study authors for additional details on partially reported outcomes.

A limitation of this review is that the evidence reported was based on a single study of 100 adults, making it impossible to assess the presence of publication bias, and leading to imprecision and reduced certainty of evidence. Additionally, given the small number of included studies, meta-analysis was not possible, thereby limiting the robustness of conclusions regarding the effectiveness, safety, and cost-effectiveness of TKA compared to non-surgical interventions of the knee for adults with moderate to severe osteoarthritis.

Agreements and disagreements with other studies or reviews

This is the first systematic review of RCTs comparing TKA to non-surgical treatments for moderate to severe osteoarthritis. Our findings align with existing literature demonstrating that TKA can provide pain relief and improve physical function in adults with primary osteoarthritis [30, 117, 118]. Our review also highlighted that there may be no difference in quality of life at one-year between adults undergoing TKA and those receiving non-surgical treatments. This finding contrasts with a meta-analysis of observational studies that demonstrated improved quality of life with TKA [30]. The discrepancies between findings could be attributed to differences in study designs, patient populations, or outcome measures used, and further research is needed to clarify these differences in outcomes. Finally, similar to our findings, prior studies have similarly reported that TKA is associated with higher risks of complications (e.g. venous thromboembolism, deep infection, cerebrovascular accident, pneumonia) [22, 33].

While some studies have suggested the cost-effectiveness of TKA [119, 120, 121], the study included in our review, which was conducted in Denmark, found that TKA followed by a non-surgical programme may not be more cost-effective than a non-surgical programme alone. Skou 2015 reported a high incremental cost-effectiveness ratio at two years (23.2%), indicating that the additional cost of TKA followed by a non-surgical programme exceeded its additional health benefit by 23.2%. These findings are consistent with another cost-effectiveness analysis, which concluded that TKA is not cost-effective when compared to non-surgical management [122]. However, the same study suggested that TKA could become cost-effective if primarily offered to adults with more severe osteoarthritis [122].

AUTHORS' CONCLUSIONS

Implications for practice

This systematic review highlights that for adults with mild to severe knee osteoarthritis, low-certainty evidence suggests that total knee arthroplasty combined with a non-surgical programme (compared to the same non-surgical programme alone) may improve pain at a level that is clinically important; may improve physical function at a level that is not clinically important; and may reduce the need for a follow-up knee surgery (revision or subsequent knee surgery and initial knee surgery). Moderate-certainty evidence shows there is likely no clinically important difference in health-related quality of life between total knee arthroplasty followed by a non-surgical programme and a non-surgical programme alone. Furthermore, total knee arthroplasty followed by a non-surgical programme may increase the risk of serious adverse events, and there may be no difference in withdrawals due to adverse events, but the evidence for both of these outcomes is very uncertain.

The continued improvement in pain and physical function over two years in the non-surgical group suggests that many patients benefited from conservative management, even if some ultimately underwent surgery. While total knee arthroplasty improves pain and physical function, it carries risks, and not all patients are willing to accept those risks. The intensive non-surgical programme used in the included study [103] reflects best-practice guidelines and serves as a strong example of what non-surgical treatments can achieve when implemented systematically. While such programmes may not be widely available, the outcomes highlight the potential of non-surgical strategies when supported by policy and infrastructure.

Approximately 25% of adults in the non-surgical group opted for total knee arthroplasty within two years, likely reflecting the natural progression of osteoarthritis. This emphasises the importance of offering a spectrum of treatment options that accommodate individual needs and evolving clinical circumstances. Clinicians and adults should carefully weigh the potential benefits and harms of total knee arthroplasty and other non-surgical options, particularly for adults with less severe symptoms or factors that may increase surgical risks [123]. A patient-centred approach to decision-making, considering individual values, preferences, and goals, is essential to a resulting patient-centred and quality decision [124]. However, the conclusions of this study should be interpreted with caution due to several limitations: the evidence is based on a single study conducted in Denmark between 2011 and 2013; total knee arthroplasty eligibility criteria used by orthopaedic surgeons were not clearly reported; a subset of participants (12%) had milder osteoarthritis; adults who reported severe pain in the previous week were excluded; and the included study employed a broad definition of serious adverse events. These factors may affect the reliability and generalisability of the findings.

Equity-related implications for practice

The generalisability of the findings is limited by the characteristics of the included study, which predominantly included white women with a mean age of 66 years who were overweight with mild to severe osteoarthritis in Denmark. These findings may therefore not be generalisable to more diverse populations, including younger adults, men, or those from different ethnic and cultural backgrounds (e.g. worse access to total knee arthroplasty

for Black adults in the United States) [125]. Clinicians should recognise disparities in access to non-surgical interventions for knee osteoarthritis and to total knee arthroplasty [92, 93, 94], and also the variability in preoperative expectations and postoperative outcomes based on sex, ethnicity, and socioeconomic factors that can influence health decisions [95, 96, 97]. Strategies should prioritise personalised approaches to address these disparities, particularly by ensuring equitable access to both surgical and non-surgical interventions for all demographic groups. Importantly, future studies that include diverse samples can inform more inclusive practice guidelines and address equity gaps in treatment.

Implications for research

Future research should focus on large, high-quality randomised controlled trials comparing total knee arthroplasty to non-surgical interventions for knee osteoarthritis. These studies should address the short-, intermediate-, and long-term benefits, harms, and cost-effectiveness of total knee arthroplasty compared to non-surgical intervention for knee osteoarthritis across diverse populations and healthcare settings. Additionally, these studies should compare treatment outcomes for moderate osteoarthritis versus severe osteoarthritis. Evidence is also needed on patient satisfaction with treatment outcomes, and partial knee arthroplasty compared to non-surgical interventions. Given the concerns about the broad definition and classification of serious adverse events in the included study, future research should use standardised and clinically meaningful criteria for reporting serious adverse events. The use of validated classification systems, such as the Clavien-Dindo system [116], could improve consistency and allow for a more accurate assessment of the risks of total knee arthroplasty compared to non-surgical interventions for the knee.

Equity-related implications for research

Following the PROGRESS-Plus framework (Place of residence, Race/ethnicity/culture/language, Occupation, Gender/sex, Religion, Education, Socioeconomic status, Social capital – Personal characteristics, features of relationships, time-dependent relationships), which supports an equity perspective in research [126], future research should prioritise subgroup analyses based on age, sex/gender, ethnicity, other demographic factors (e.g. place of residence, language, occupation, religion, education, socioeconomic status, social capital), and other factors (e.g. personal characteristics associated with discrimination) to better understand differential outcomes of total knee arthroplasty and non-surgical interventions. These subgroups are particularly relevant due to known disparities in the prevalence of knee osteoarthritis [91], access to total knee arthroplasty [92, 93, 94], access to quality and free non-surgical treatments for knee osteoarthritis [127, 128], and postoperative outcomes among diverse populations [95, 96, 97]. Studies should aim to include adults from under-represented groups to improve the applicability of findings. By incorporating diverse sample characteristics, researchers can address equity-related implications for both practice and policy. Additionally, research should explore the influence of a variety of non-surgical intervention types, implant types, and ideally placebo/sham on outcomes. Finally, expanding the geographic and cultural diversity of study settings will further enhance the relevance and equity of research findings to a broader population.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD015378.pub2](https://doi.org/10.1002/14651858.CD015378.pub2).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of ongoing studies

Supplementary material 5 Analyses

Supplementary material 6 Data package

Supplementary material 7 Main characteristics of participants

Supplementary material 8 Outcomes reported by studies

ADDITIONAL INFORMATION

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Contributions of authors

LPB screened articles, collected and analysed the data, wrote the manuscript, contributed to the decisions and development of the manuscript, and approved the final manuscript.

SA screened articles and collected data, contributed to the decisions and development of the manuscript, and approved the final manuscript.

SK resolved any disagreement in screening, contributed to the decisions and development of the manuscript, and approved the final manuscript.

JPP developed and conducted the search strategy, resolved any disagreement in screening, contributed to the decisions and development of the manuscript, and approved the final manuscript.

GD resolved any disagreement in screening, contributed to the decisions and development of the manuscript, and approved the final manuscript.

DS resolved any disagreement in screening, contributed to the decisions and development of the manuscript, and approved the final manuscript.

GW resolved any disagreement in screening, contributed to the decisions and development of the manuscript, and approved the final manuscript.

Declarations of interest

LPB has no conflict of interest to declare.

SA has no conflict of interest to declare.

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JPP has a Cochrane editorial board (member) and thematic group role (Senior Advisor, Campbell and Cochrane Health Equity Thematic Group). JPP was not involved in the editorial process of this review.

GD has previously served as a consultant for Microport Orthopedics and has received research funding from the company to support unrelated studies on implant design longevity and the effect of different implant designs on kinematics. These studies were not eligible for inclusion in the current review because they did not meet the inclusion criteria. GD also received honoraria for speaking engagements from Zimmer Biomet. Neither company had any involvement in the design, conduct, analysis, or reporting of this study.

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Data, code and other materials

As part of the published Cochrane review, the following is made available for download for users of the Cochrane Library (see [Supplementary material 6](#)): full search strategies for each database ([Supplementary material 1](#)): full citations of each unique report for all studies (included, ongoing, or excluded at the full text screen) in the final review; study data, including study information, study arms, and study results; consensus risk of bias assessments; and analysis data. The template data extraction form from Excel is available from the authors on reasonable request..

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ADDITIONAL TABLES

Table 1. Overview of Syntheses and Included Studies

Study	Country	Design	Population	Intervention	Comparison	Follow-up
Skou 2015	Denmark	RCT 2 arms	Primary osteoarthritis	TKA combined with a 12-week non-surgical programme: TKA: total cemented prosthesis with patellar resurfacing (NexGen CR-Flex or LPS-Flex Fixed Bear-	12-week non-surgical programme: - Exercises by physiotherapist: neuromuscular exercise training programme (1-hour group session, 2 times/week, 12 weeks) - Education by physiotherapist: disease characteristics, treatments, and self-management strategies (2 sessions, 1 hour) - Dietary advice by dietitian: weight loss programme for participants with BMI ≥ 25 (4 sessions, 30 to 60 minutes/session, 30-minute telephone follow-up at weeks 26 and 39)	3 months 6 months 12 months 2 years

Table 1. Overview of Syntheses and Included Studies *(Continued)*

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|--|---|
| ing Knee,
Zimmer*) | <ul style="list-style-type: none"> • Insoles: individualised insoles with medial arch support and, when needed, 4-degree lateral wedge |
| <ul style="list-style-type: none"> • Non-surgical programme: see comparison section | <ul style="list-style-type: none"> • Pain medication by surgeon (if adequate/needed): as needed, paracetamol (1 g, 4 times/day), ibuprofen (400 mg, 3 times/day), pantoprazole (20 mg/day) • Adherence: physiotherapist contacted participants monthly by telephone until the 12-month follow-up assessment |

BMI: body mass index, RCT: randomised controlled trial, TKA: total knee arthroplasty.

*The LPS-Flex and NexGen CR-Flex, from the company Zimmer Biomet, are the two types of knee implants used to replace the knee joint of the study participants.

INDEX TERMS

Medical Subject Headings (MeSH)

Aged; *Arthroplasty, Replacement, Knee [adverse effects] [methods]; Bias; Braces; Humans; Middle Aged; Orthotic Devices;

*Osteoarthritis, Knee [surgery] [therapy]; Quality of Life; Randomized Controlled Trials as Topic